
Open economy HANK

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Open economy HANK

- ❖ So far, focus on *closed* economy models of fiscal & monetary policy
- ❖ **Next:** *Open* economy. What changes?
 - ❖ Exports & imports are new **source** and **destination** for demand
 - ❖ Extent to which is controlled by the **exchange rate**
- ❖ Material here based on Gali Monacelli (2005) and Auclert, Rognlie, Souchier, Straub (2024)

Exciting other work in this area: de Ferra et al (2020), Cugat (2019), Giagheddu (2020), Zhou (2022), Kekre Lenel (2020), Guo Ottonello Perez (2021), Aggarwal et al (2022), Sundry (2024), Bellifemine Couturier Jamilov (2025, R&R REStud)

Proceed in three steps

1. Introduce model that nests RA & HA
 - ❖ RA model almost literally = Gali Monacelli (2005)
 - ❖ HA model: no bonds, but capitalized profits
 - ❖ Key parameter: **trade elasticity** χ
2. Effects of **exchange rate shocks** (e.g. due to capital flows or UIP shocks)
3. Paper: Effects of **monetary policy**

HANK meets Gali-Monacelli

Model overview

- ❖ Small open economy (SOE) model
- ❖ Two goods
 - ❖ “Home”: H , produced at home, P_{Ht} at home, P_{Ht}^* abroad
 - ❖ “Foreign”: F , produced abroad, P_{Ft} at home, $P_{Ft}^* \equiv 1$ abroad
 - ❖ Consumed in bundles. CPI P_t at home, P_t^* abroad
- ❖ Two kinds of agents:
 - ❖ Large mass of foreign households
 - ❖ mass 1 of **HA domestic households**

Households' consumption behavior

- ❖ Foreigners consume fixed real C^* . Home HA solve **intertemporal problem**:

$$\max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_{it} \left(u(c_{it}) - v(N_t) \right) \quad c_{it} + a_{it} \leq (1 + r_t^p) a_{it-1} + \underbrace{Z_t e_{it}}_{\text{real labor income}} \quad a_{it} \geq 0$$

- ❖ Domestic & foreign consume CES bundle, solve **intratemporal problem**:

$$C_{Ht} = (1 - \alpha) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} C_t \quad C_{Ht}^* = \alpha \left(\frac{P_{Ht}^*}{P^*} \right)^{-\gamma} C^*$$

- ❖ Domestic production and market clearing: $Y_t = N_t = C_{Ht} + C_{Ht}^*$

Prices and nominal rigidities

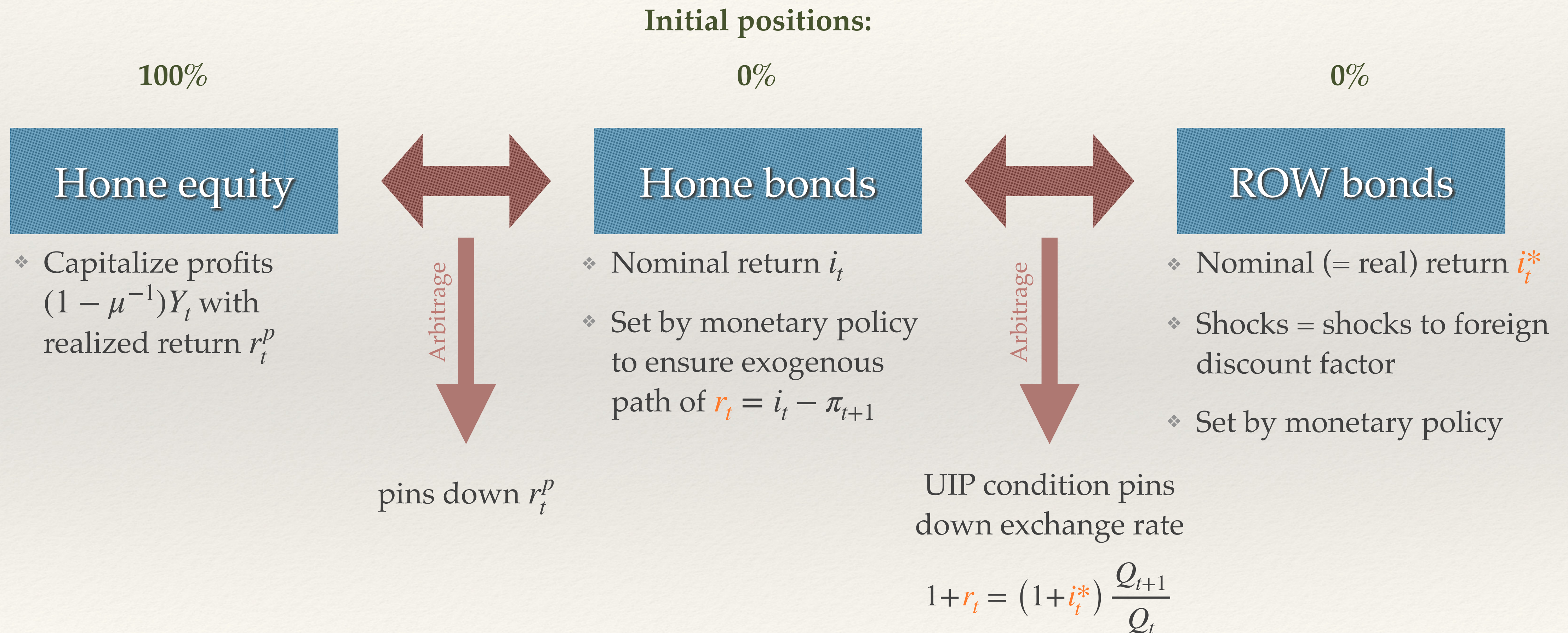
- ❖ Exchange rates: nominal $\mathcal{E}_t$, real $Q_t \equiv \mathcal{E}_t/P_t$, $\uparrow$ is depreciation
- ❖ Same wage rigidity as before

$$\pi_{wt} = \kappa_w \left(v'(N_t) - \frac{\epsilon - 1}{\epsilon} \frac{W_t}{P_t} u'(C_t) \right) + \beta \pi_{wt+1}$$

- ❖ Flexible prices everywhere else:

$$P_{Ft} = \mathcal{E}_t \quad P_{Ht} = W_t \quad P_{Ht}^* = \frac{P_{Ht}}{\mathcal{E}_t} \longleftarrow \text{“Producer currency pricing”}$$

Monetary policy and assets



Baseline calibration

- ❖ Calibrate openness $\alpha = 0.40$ & balanced trade in steady state
- ❖ Same HA block as before
- ❖ Normalize all prices to 1 in steady state.
- ❖ *Note:* HA model already stationary, no need for debt-elastic interest rate
- ❖ Next: i_t^* shocks, then (briefly) r_t shocks.

Capital flows and exchange rates

Shock

- ❖ Temporary shock i_t^* $\uparrow$
 - ❖ Real depreciation! Iterate UIP forward:

$$dQ_t = \frac{1}{1+r} \sum_{s \geq 0} di_{t+s}^*$$

- ❖ $Q_t \uparrow$ $\frac{P_{Ht}}{P_t} \downarrow$ $\frac{P_{Ht}}{\mathcal{E}_t} \downarrow$
- ❖ Demand for home goods?
- ❖ First **RA**, then **HA**

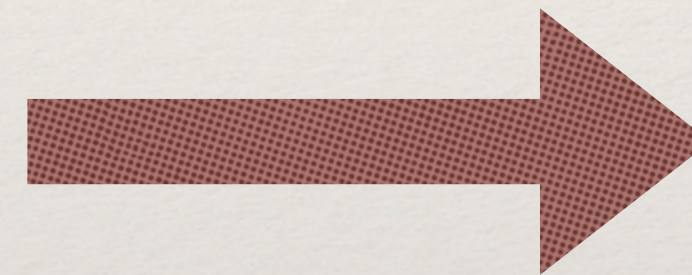
What happens to aggregate demand?

RA: $C_t = C = \text{const}!$

$$Y_t = (1 - \alpha) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} C_t + \alpha \left(\frac{P_{Ht}}{\mathcal{E}_t} \right)^{-\gamma} C^*$$

$\uparrow$ with elasticity $\eta \frac{\alpha}{1 - \alpha}$ $\uparrow$ with elasticity $\gamma \frac{1}{1 - \alpha}$

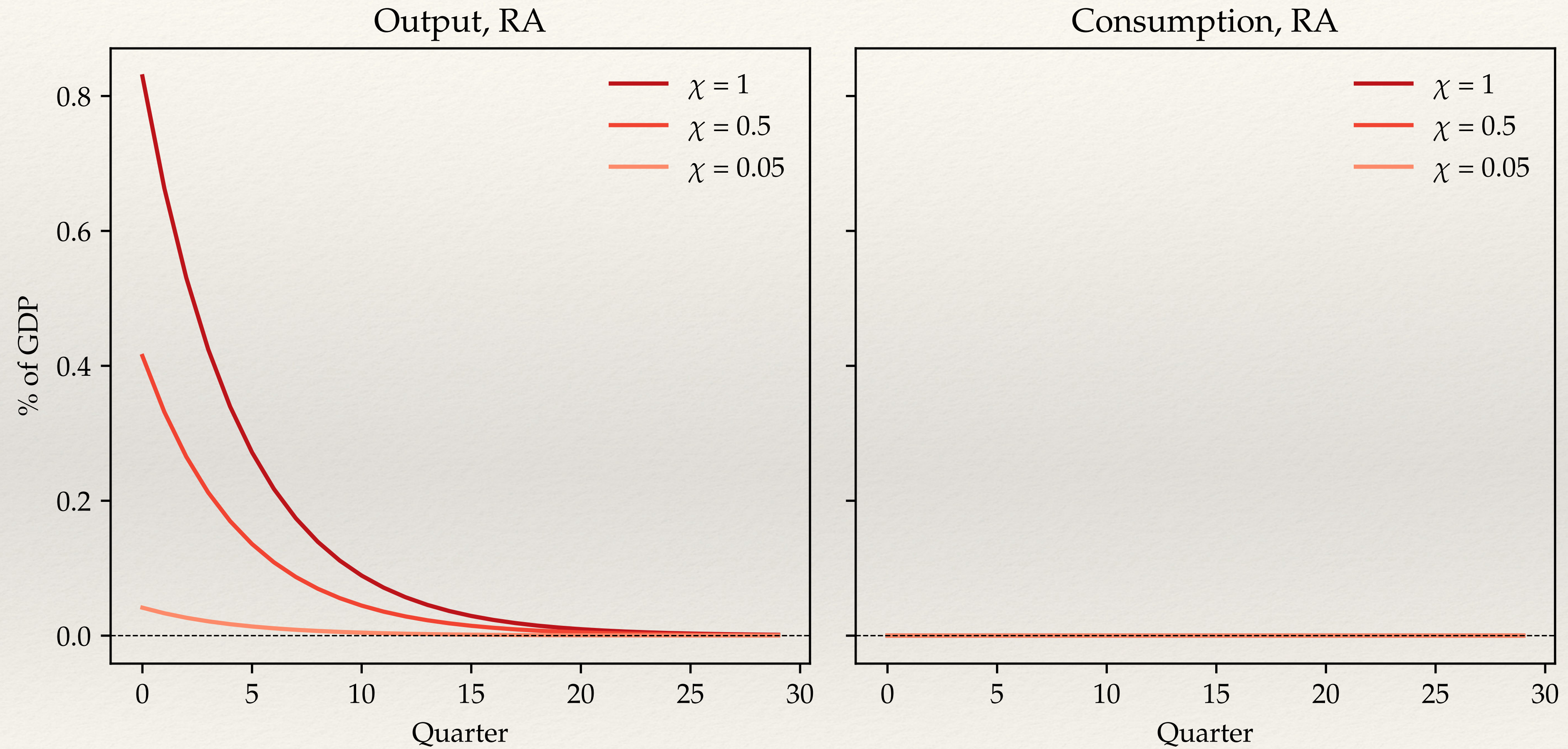
“Expenditure switching”



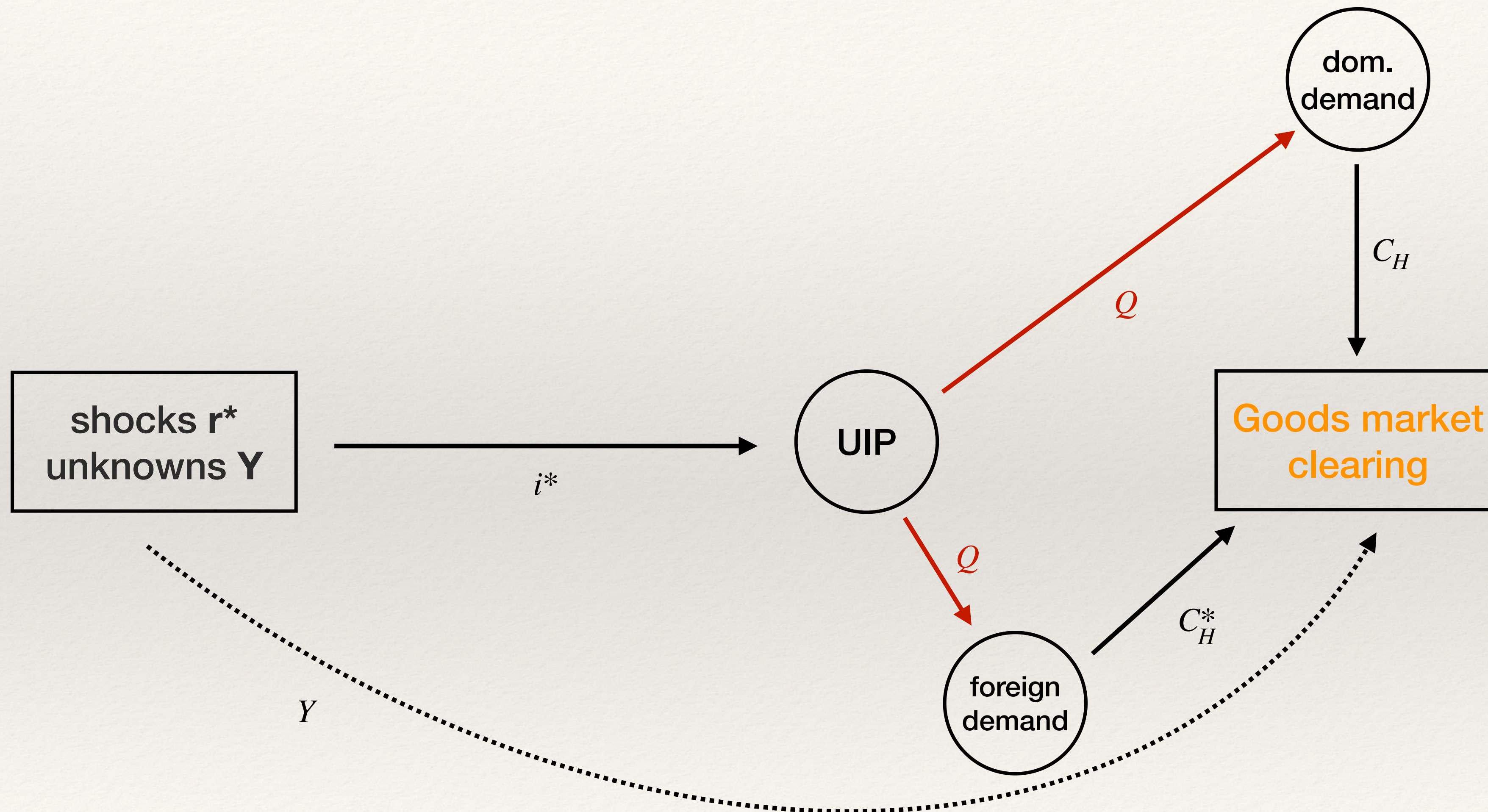
$$dY = \frac{\alpha}{1 - \alpha} \chi dQ$$

trade elasticity: $\chi \equiv \eta(1 - \alpha) + \gamma$

Representative agent: Exchange rate shock



DAG




```

@sj.solved(unknowns={'Q': (0.01, 300.)}, targets=['uip'])
def UIP(Q, r, rstar, eta, alpha, gamma):
    # recursive equation for UIP to pin down RER Q
    uip = 1 + r - (1 + rstar) * Q(1) / Q

    # price of H goods abroad in terms of Q
    PHstar = ((Q ** (eta - 1) - alpha) / (1 - alpha)) ** (1 / (1 - eta))

    # price of H goods at home in terms of Q
    PH_P = ((1 - alpha * Q ** (1 - eta)) / (1 - alpha)) ** (1 / (1 - eta))

    # price of F goods at home in terms of Q
    PF_P = Q # LOOP

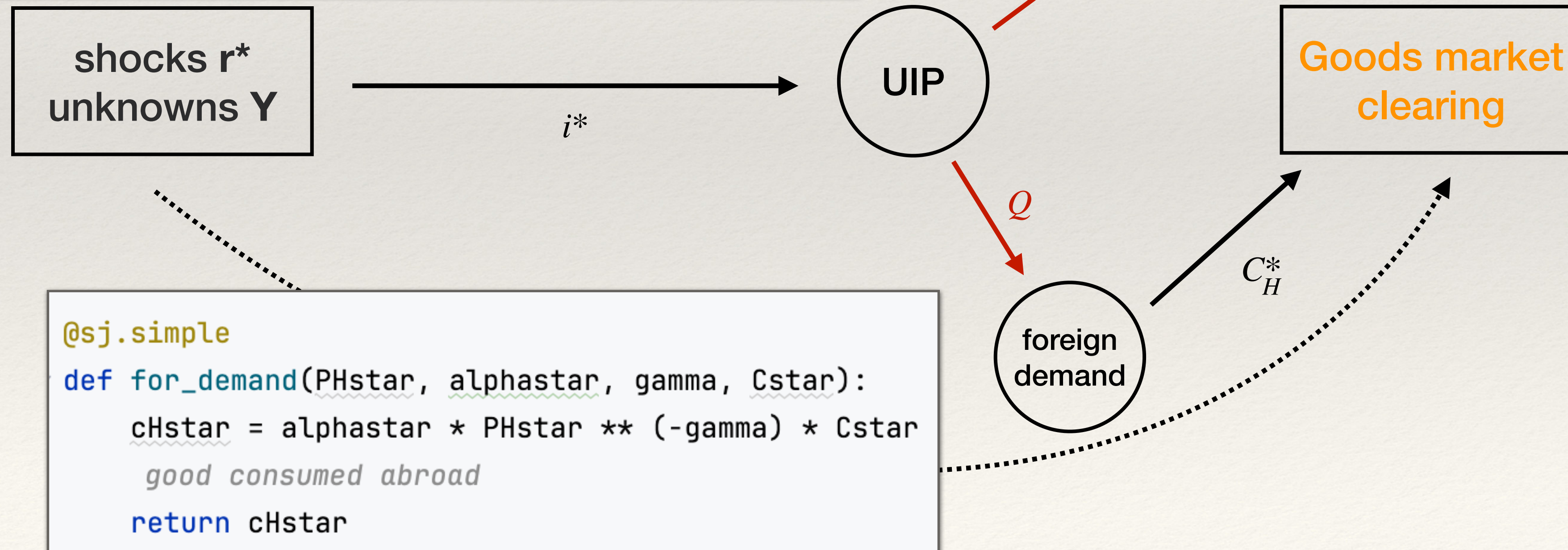
    # let's also compute chi, as an important object in the theory
    chi = eta * (1 - alpha) + gamma
    return uip, PHstar, PH_P, PF_P, chi

```

```

@sj.simple
def dom_demand(C, PF_P, PH_P, eta, alpha):
    cH = (1 - alpha) * PH_P ** (-eta) * C
    # domestically
    cF = alpha * PF_P ** (-eta) * C # PF_
    # domestically
    return cH, cF

```



What changes with heterogeneous agents?

$$\text{HA: } C_t = \mathcal{C}_t \left(r_0^p, \{Z_s\} \right)! \Rightarrow dC = \mathbf{M}/\mu d \left(\frac{P_{Ht}}{P_t} Y_t \right) + \mathbf{m} A_{ss} dr_0^p = \bar{\mathbf{M}} d \left(\frac{P_{Ht}}{P_t} Y_t \right) = -\frac{\alpha}{1-\alpha} \bar{\mathbf{M}} d\mathbf{Q} + \bar{\mathbf{M}} d\mathbf{Y}$$

$$\bar{\mathbf{M}} \equiv \frac{1}{\mu} \mathbf{M} + \left(1 - \frac{1}{\mu} \right) \mathbf{m} \mathbf{q}'$$

Intertemporal MPCs out of total income

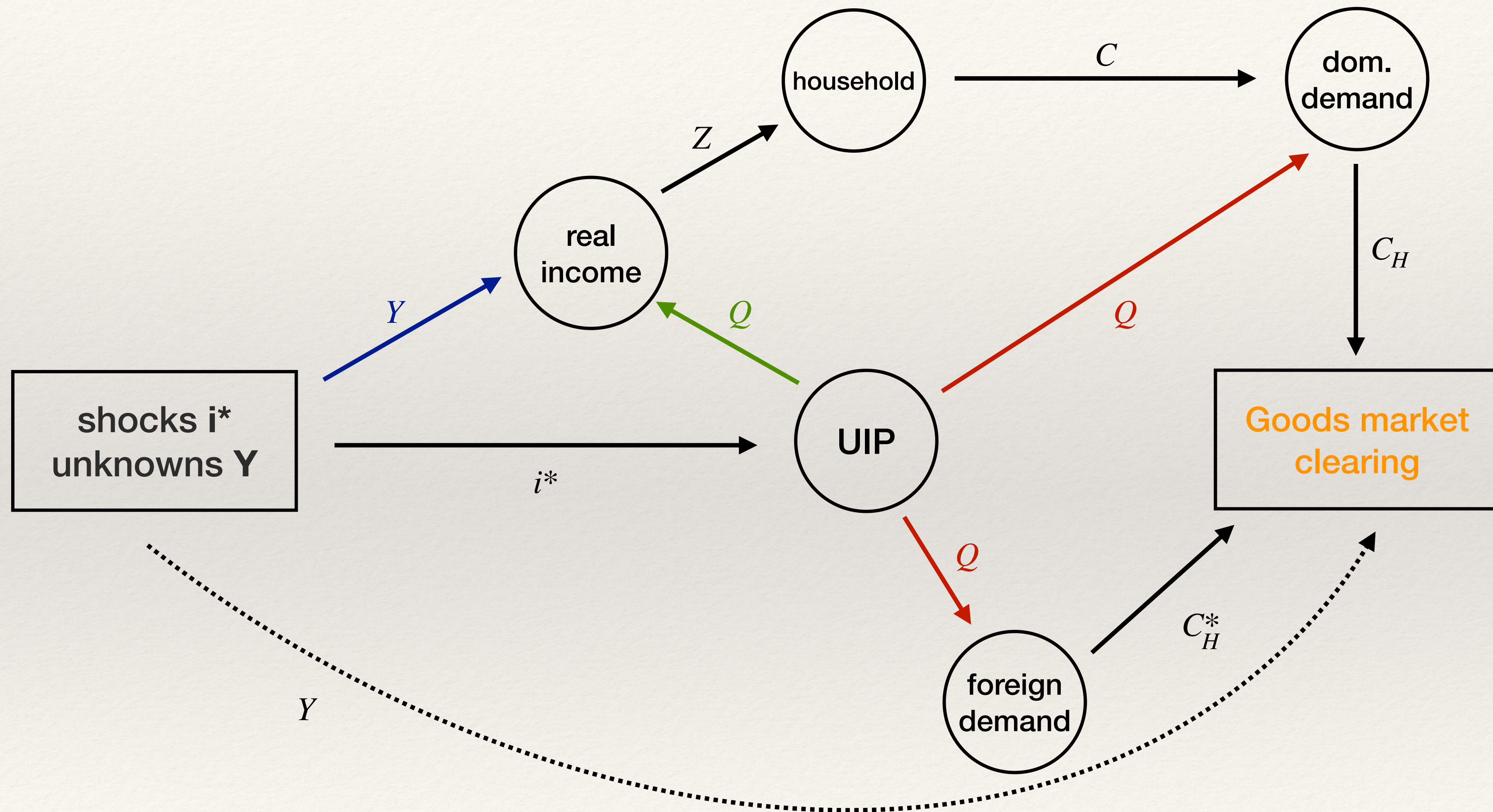
$$Y_t = (1 - \alpha) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} C_t + \alpha \left(\frac{P_{Ht}}{\mathcal{E}_t} \right)^{-\gamma} C^* \Rightarrow dY = \frac{\alpha}{1-\alpha} \chi d\mathbf{Q} - \alpha \bar{\mathbf{M}} d\mathbf{Q} + (1 - \alpha) \bar{\mathbf{M}} d\mathbf{Y}$$

Real income channel
Multiplier

Expenditure switching

↑ with elasticity $\eta(1 - \alpha)$
↑ with elasticity γ

DAG




```
@sj.solved(unknowns={'J': (0.001, 100.)}, targets=['valuation_cond'])
def income(Y, PH_P, J, r, markup_ss):
    # real labor income
    Z = 1 / markup_ss * PH_P * Y

    # real dividend
    div = (1 - 1 / markup_ss) * PH_P * Y

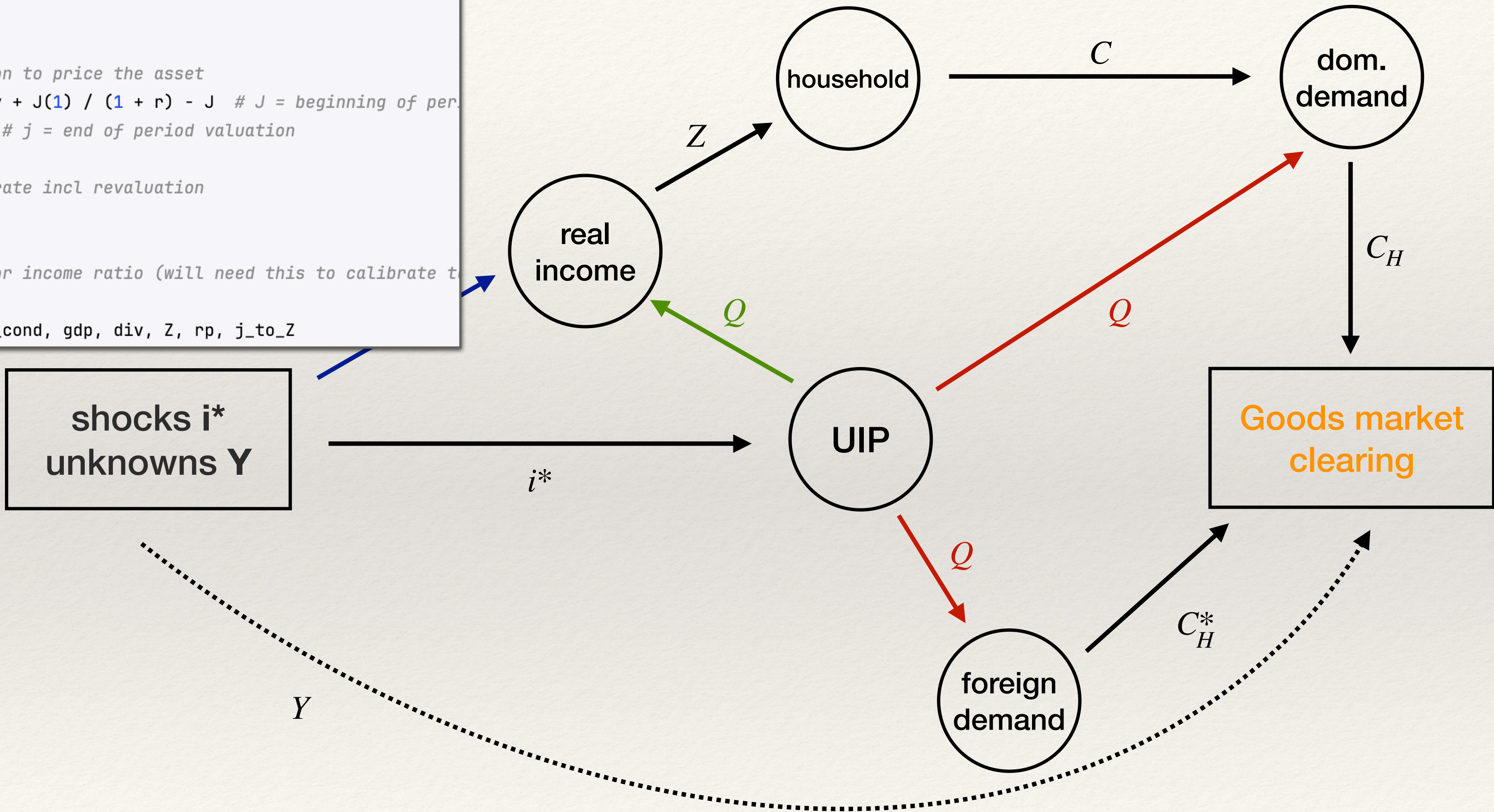
    # nominal PPP adjusted GDP
    gdp = PH_P * Y

    # valuation condition to price the asset
    valuation_cond = div + J(1) / (1 + r) - J # J = beginning of period
    j = J(1) / (1 + r) # j = end of period valuation

    # ex post interest rate incl revaluation
    rp = J / j(-1) - 1

    # get assets to labor income ratio (will need this to calibrate to data)
    j_to_Z = j / Z
    return j, valuation_cond, gdp, div, Z, rp, j_to_Z
```

DAG



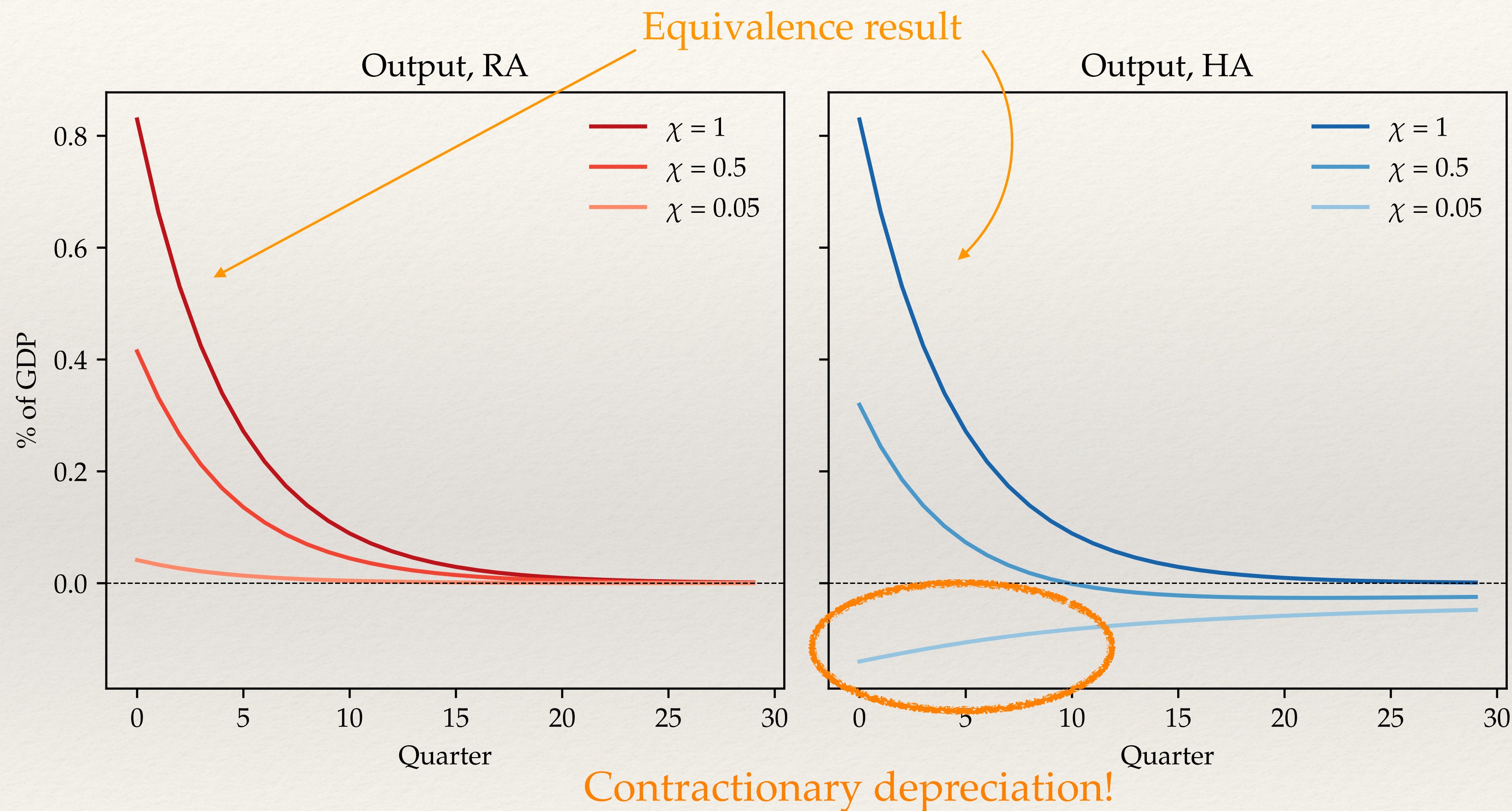
How do RA and HA compare?

- ❖ Assume $\chi = 1$. Then: $d\mathbf{Y}^{HA} = d\mathbf{Y}^{RA} = \frac{\alpha}{1 - \alpha} d\mathbf{Q}$
- ❖ HA and RA are identical in this case! What about the two new terms? **Cancel!**

$$\alpha \mathbf{M} d\mathbf{Q} = (1 - \alpha) \mathbf{M} d\mathbf{Y}$$

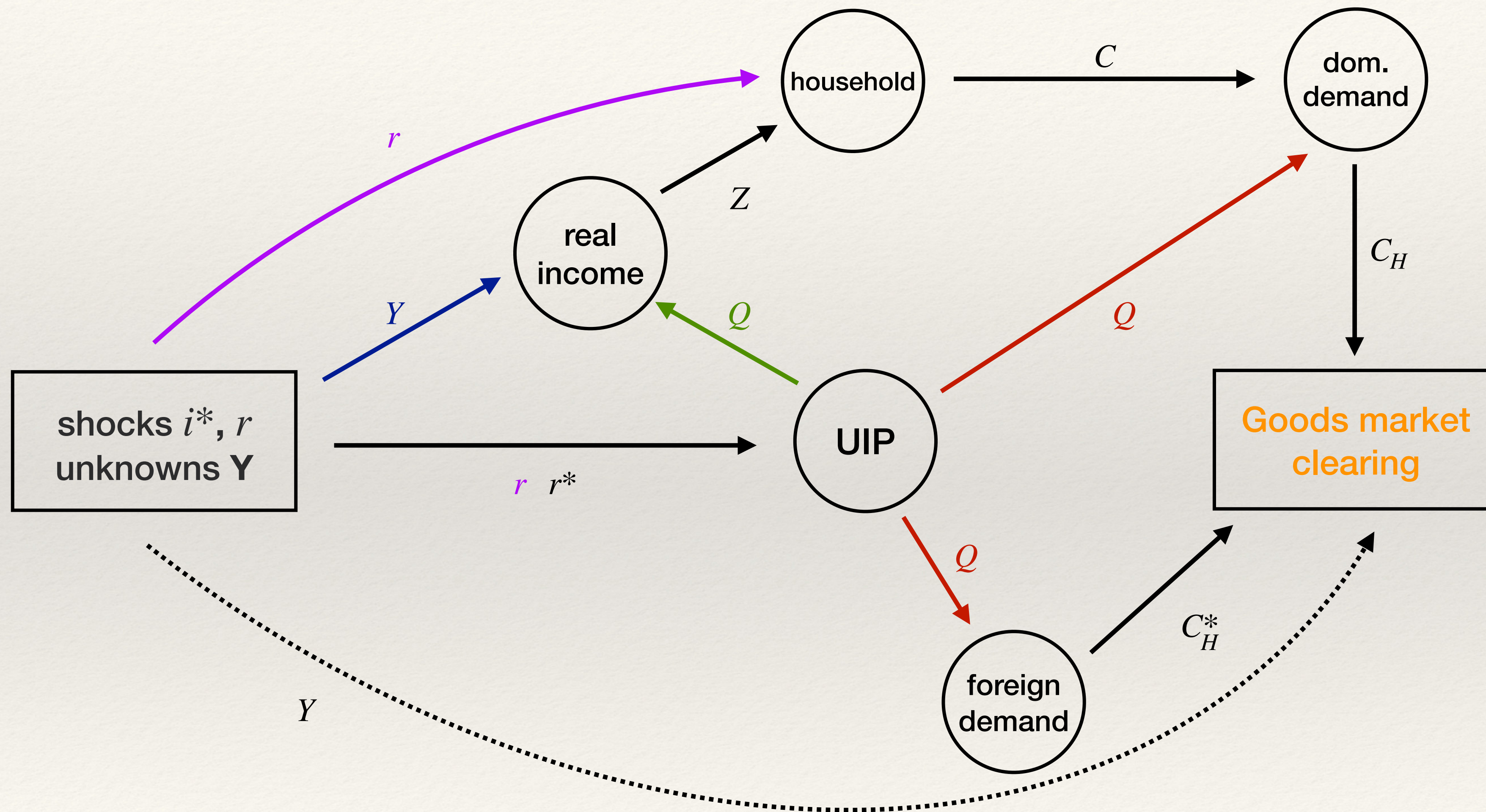
- ❖ **Intuition:** Depreciation causes just enough of a boom that the loss in real income due to depreciation is offset. [geeky comment: this is a little like the balanced budget multiplier]
- ❖ What if $\chi \neq 1$?

Contractionary depreciations for low χ



- ❖ This is more likely when substitution away from imports is hard (e.g. energy, food)

What about monetary policy?



Summary

- ❖ Merged HANK with Gali-Monacelli.
 - ❖ Maybe the most natural way to apply HANK to open economies?
- ❖ Learned:
 - ❖ New channels: **Real income**, **Keynesian Multiplier**
 - ❖ Can generate contractionary depreciations for low trade elasticities
- ❖ Lots more in paper: Taylor rules, non-trivial gross positions, slow trade adjustments (J curve), non-homothetic demand, DCP, slow pass-through, ...